Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	28 October 2026
Team ID	EN25CS3010040
Project Name	Global Olympic Sports Performance Analysis
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyze data.	Python, SQL, HTML, CSS, etc.	Supports data queries, calculations and analytical operations required for Olympic performance analysis.
2.	Database / Data Warehouse	Used to store and manage structured business data for analysis and reporting.	Snowflake / MySQL / Excel / CSV	Stores the four institutional CSV datasets used as the project's source data.
3.	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Snowflake Stages / COPY INTO / Tableau Data Connections	Connects the source CSV files to Tableau for analysis and visualization.
4.	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	Eg. Tableau Prep, Snowflake SQL, Tableau Desktop	Enables data relationships, field configuration, calculated measures and filter setup in Tableau.
5.	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Snowflake SQL Charts, Tableau, Python	Creates interactive charts, three dashboards and a four-point Tableau Story to communicate Olympic performance insights.
6.	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	Tableau Public, Flask, Streamlit	Allows users to view published dashboards and interact with supported filters through Tableau Public.